

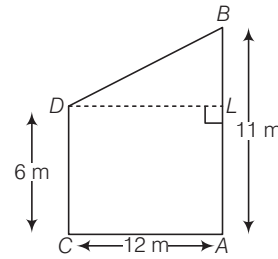
Converse of Pythagoras theorem

In a triangle, if the square of one side is equal to the sum of square of the other two sides then angle opposite to the first side is a right angle i.e. In $\triangle ABC$, if $AC^2 = AB^2 + BC^2$, then $\angle B = 90^\circ$.

EXAMPLE 12. Two poles of heights 6m and 11m stand vertically on a plane ground at a distance of 12m from each other. The distance between their tops is

- a. 9 m b. 10 m
c. 13 m d. 15 m

Sol. c. Let AB and CD be two poles such that $AB = 11\text{m}$, $CD = 6\text{ m}$ and $AC = 12\text{ m}$.



Draw $DL \perp AB$

Then, $BL = AB - AL = AB - CD = (11 - 6)\text{m} = 5\text{m}$

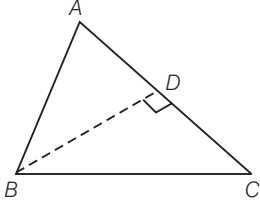
and $DL = AC = 12\text{m}$

In right angled $\triangle BDL$,

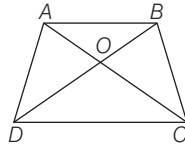
$$BD^2 = DL^2 + BL^2 = 5^2 + 12^2 = 169$$

$$\Rightarrow BD = \sqrt{169} = 13\text{m}$$

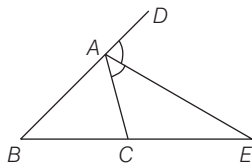
> PRACTICE EXERCISE

- If the bisector of an angle of a triangle bisects the opposite side, then the triangle is
(a) equilateral (b) isosceles
(c) scalene (d) right angled triangle
- The line segments joining the mid-points of the sides of a triangle form four triangles each of which is
(a) similar to the original triangle
(b) congruent to the original triangle
(c) an equilateral triangle (d) an isosceles triangle
- In a $\triangle ABC$, BD and CE are perpendicular on AC and AB , respectively. If $BD = CE$, then the $\triangle ABC$ is
(a) equilateral (b) isosceles (c) right angled (d) scalene
- If the length of hypotenuse of a right angled triangle is 5 cm and its area is 6 cm^2 , then the length of the remaining sides are
(a) 1 cm and 3 cm (b) 3 cm and 2 cm
(c) 3 cm and 4 cm (d) 4 cm and 2 cm
- $\triangle ABC$ is such that $AB = 3\text{ cm}$, $BC = 2\text{ cm}$ and $AC = 2.5\text{ cm}$. $\triangle DEF$ is similar to $\triangle ABC$. If $EF = 4\text{ cm}$, then the perimeter of $\triangle DEF$ is
(a) 5 cm (b) 7.5 cm (c) 15 cm (d) 18 cm
- A soldier goes to a warfield and runs in the following manner. From the starting point, he goes West 25 m, then due North 60 m, then due East 80 m, and finally due South 12 cm. The distance between the starting point and the finishing point is
(a) 177 m (b) 103 m (c) 83 m (d) 73 m
- Let ABC be an isosceles triangle in which $AB = AC$ and $BD \perp AC$. Then, $BD^2 - CD^2$ is equal to

(a) $2DC \cdot AD$ (b) $2AD \cdot BC$ (c) $3DC \cdot AD$ (d) $\frac{1}{2} AD \cdot DC$
- D and E are the points on the sides AB and AC respectively of a $\triangle ABC$ and $AD = 8\text{ cm}$, $DB = 12\text{ cm}$, $AE = 6\text{ cm}$ and $EC = 9\text{ cm}$, then BC is equal to
(a) $\frac{2}{5} DE$ (b) $\frac{5}{2} DE$ (c) $\frac{3}{2} DE$ (d) $\frac{2}{3} DE$
- A vertical stick 15m long casts a shadow 12m long on the ground. At the same time, a tower casts a shadow 50m long on the ground. The height of the tower is
(a) 60 m (b) 62 m (c) 62.5 m (d) 63 m
- The areas of two similar triangles are 81 cm^2 and 49 cm^2 , respectively. The ratio of their corresponding heights is
(a) 9 : 7 (b) 7 : 9 (c) 6 : 5 (d) 81 : 49
- If D and E are points on the sides AB and AC , respectively of a $\triangle ABC$ such that $DE \parallel BC$. If $AD = x$, $DB = x - 2$, $AE = x + 2$ and $EC = x - 1$. The value of x is
(a) 2.5 (b) 2 (c) 3 (d) 4

- 12.** In the adjoining figure, $ABCD$ is a trapezium in which $AB \parallel CD$ and its diagonals intersect at O . If $AO = (3x - 1)$, $OC = (5x - 3)$, $BO = (2x + 1)$ and $OD = (6x - 5)$, then x is equal to



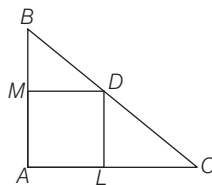
- (a) 1 (b) 2 (c) 3 (d) 4
- 13.** In the adjoining figure, AE is the bisector of exterior $\angle CAD$ meeting BC produced in E . If $AB = 10$ cm, $AC = 6$ cm and $BC = 12$ cm, then CE is equal to



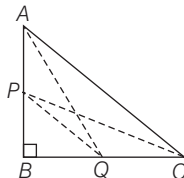
- (a) 6 cm (b) 12 cm (c) 18 cm (d) 20 cm
- 14.** If D, E and F are respectively the mid-points of sides BC, AC and AB of a $\triangle ABC$. If $EF = 3$ cm, $FD = 4$ cm and $AB = 10$ cm, then DE, BC and CA , respectively will be equal to

- (a) 6, 8 and 20 cm (b) $\frac{10}{3}$, 9 and 12 cm
(c) 4, 6 and 8 cm (d) 5, 6 and 8 cm
- 15.** In $\triangle PQR$, $\angle Q = 3a$, $\angle P = a$, $\angle R = b$ and $3b - 5a = 30$, then the triangle is
- (a) scalene (b) isosceles
(c) equilateral (d) right angled

- 16.** In $\triangle ABC$ show in the figure $\angle A = 90^\circ$. Let D be a point on BC such that $BD : DC = 1 : 3$. If DM and DL , respectively are perpendicular on AB and AC , then DM and LC are in the ratio of

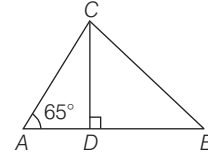


- (a) 1 : 3 (b) 1 : 2 (c) 1 : 1 (d) 4 : 1
- 17.** In a right angled $\triangle ABC$, right angled at B , if P and Q are points on the sides AB and AC respectively, then

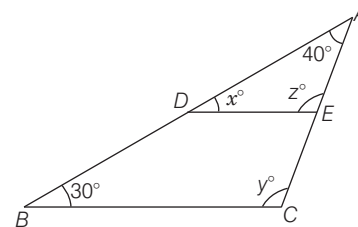


- (a) $AQ^2 + CP^2 = 2(AC^2 + PQ^2)$
(b) $2(AQ^2 + CP^2) = AC^2 + PQ^2$
(c) $AQ^2 + CP^2 = AC^2 + PQ^2$
(d) $AQ + CP = \frac{1}{2}(AC + PQ)$

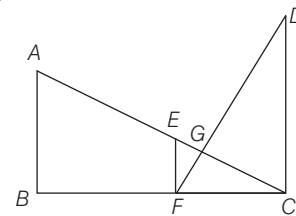
- 18.** In $\triangle ABC$, $\angle C = 90^\circ$ and $CD \perp AB$, also $\angle A = 65^\circ$, then $\angle CBA$ is equal to



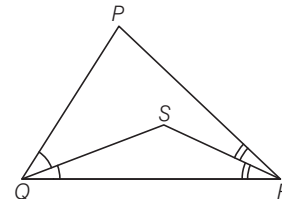
- (a) 25° (b) 35° (c) 65° (d) 40°
- 19.** The angles of a triangle are in the ratio 2 : 3 : 4. The angles of triangle are, respectively
- (a) $30^\circ, 60^\circ, 90^\circ$ (b) $40^\circ, 60^\circ, 80^\circ$
(c) $60^\circ, 40^\circ, 80^\circ$ (d) $20^\circ, 60^\circ, 80^\circ$
- 20.** In figure, D and E are points on sides AB, AC of $\triangle ABC$ such that $DE \parallel BC$. If $\angle B = 30^\circ$ and $\angle A = 40^\circ$, then x, y and z are, respectively



- (a) $30^\circ, 110^\circ, 110^\circ$ (b) $30^\circ, 105^\circ, 105^\circ$
(c) $30^\circ, 85^\circ, 85^\circ$ (d) $30^\circ, 95^\circ, 95^\circ$
- 21.** In figure, AB, EF and CD are parallel lines. Given that $GE = 5$ cm, $GC = 10$ cm and $DC = 18$ cm, then EF is equal to



- (a) 11 cm (b) 5 cm (c) 6 cm (d) 9 cm
- 22.** In the given figure, $PQ > PR$. QS and RS are the bisectors of $\angle Q$ and $\angle R$ respectively, then which of the following is correct?



- (a) $SQ < SR$ (b) $SQ = SR$
(c) $SQ > SR$ (d) None of these
- 23.** In a $\triangle ABC$, $AB = AC$ and $60^\circ < A < 90^\circ$. If $AB = c$, $AC = b$ and $BC = a$, then which of the following is correct?
- (a) $b < a < 2b$ (b) $c/3 < a < 3c$
(c) $b < a < b\sqrt{3}$ (d) $c < a < c\sqrt{2}$

24. In $\triangle ABC$, $AD \perp BC$, then

- (a) $AB^2 - BD^2 = AC^2 - CD^2$ (b) $AB^2 + BD^2 = AC^2 - CD^2$
 (c) $AB^2 + BD^2 = AC^2 + CD^2$ (d) $AB^2 + AC^2 = BD^2 + DC^2$

25. $\triangle ABC$ is a right angled at C and P is the length of the perpendicular from C to AB . If $BC = a$, $AC = b$, $AB = c$, then

- (a) $\frac{a}{b} = \frac{p}{c}$ (b) $pc = ab$ (c) $\frac{1}{a} + \frac{1}{b} = \frac{1}{ab}$ (d) None of these

26. ABC is a triangle and the perpendicular drawn from A meets BC in D . If $AD^2 = BD \cdot DC$, then which one of the following is correct?

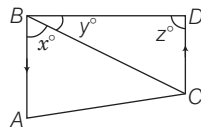
- (a) ABC must be an obtuse angled triangle
 (b) ABC must be an acute angled triangle
 (c) Either $\angle B \geq 45^\circ$ or $\angle C \geq 45^\circ$ (d) $BC^2 = AB^2 + AC^2$

27. If Δ is the area of a right angled triangle and b is one of the sides containing the right angle, then what is the length of the altitude on the hypotenuse?

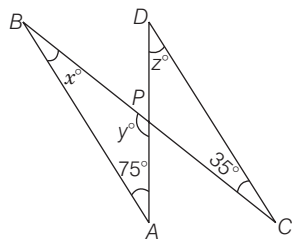
- (a) $\frac{2 \Delta b}{\sqrt{b^4 + 4\Delta^2}}$ (b) $\frac{2 \Delta^2 b}{\sqrt{b^4 + 4\Delta^2}}$
 (c) $\frac{2 \Delta b^2}{\sqrt{b^4 + 4\Delta^2}}$ (d) $\frac{2 \Delta^2 b^2}{\sqrt{b^4 + \Delta^2}}$

28. In figure, $AB \parallel CD$. If $x = \frac{4}{3}y$ and $y = \frac{3}{8}z$, then the value of x is

- (a) 48° (b) 96° (c) 36° (d) None of these

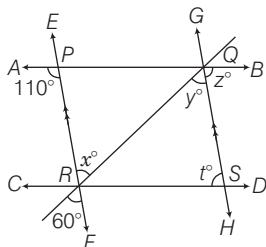


29. In the given figure, $AB \parallel CD$, then the values of x, y and z are, respectively



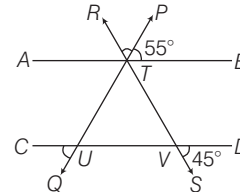
- (a) $75^\circ, 35^\circ, 80^\circ$ (b) $70^\circ, 35^\circ, 60^\circ$
 (c) $35^\circ, 70^\circ, 75^\circ$ (d) $70^\circ, 35^\circ, 80^\circ$

30. In the given figure, $AB \parallel CD$, and $EF \parallel GH$. The values of x, y, z and t are respectively



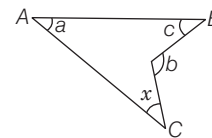
- (a) $60^\circ, 75^\circ, 75^\circ, 60^\circ$ (b) $50^\circ, 75^\circ, 75^\circ, 65^\circ$
 (c) $60^\circ, 70^\circ, 60^\circ, 70^\circ$ (d) $60^\circ, 60^\circ, 70^\circ, 70^\circ$

31. In the given figure, $AB \parallel CD$, $\angle PTB = 55^\circ$ and $\angle DVS = 45^\circ$, then what is the sum of the measures of $\angle CUQ$ and $\angle RTP$?



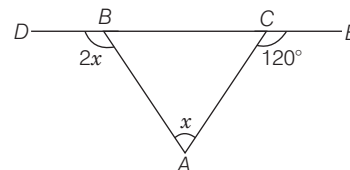
- (a) 180° (b) 135° (c) 110° (d) 100°

32. What is the value of x in the figure given below?



- (a) $b - a - c$ (b) $b - a + c$ (c) $b + a - c$ (d) $\pi - (a + b + c)$

33. In the given figure, what is the value of x ?

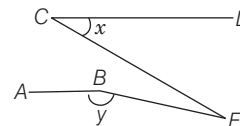


- (a) 30° (b) 40° (c) 45° (d) 60°

34. PQR is a triangle right angled at Q . If X and Y are the mid-points of the sides PQ and QR respectively, then which one of the following is not correct?

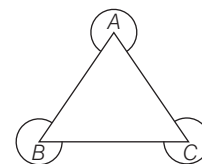
- (a) $RX^2 + PY^2 = 5XY^2$ (b) $RX^2 + PY^2 = XY^2 + PR^2$
 (c) $4(RX^2 + PY^2) = 5PR^2$ (d) $RX^2 + PY^2 = 3(PQ^2 + QR^2)$

35. In the given figure, AB is parallel to CD . If $\angle DCE = x$ and $\angle ABE = y$, then what is $\angle CEB$ equal to?



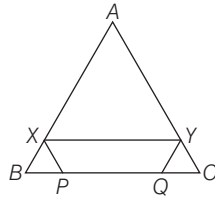
- (a) $y - x$ (b) $(x + y)/2$ (c) $x + y - (\pi/2)$ (d) $x + y - \pi$

36. In the figure given below, what is the sum of the angles formed around A, B, C except the angles of the $\triangle ABC$?

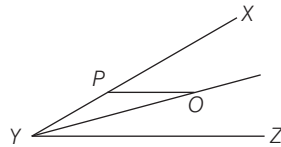


- (a) 360° (b) 720° (c) 900° (d) 1000°

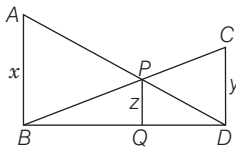
- 37.** In the given figure, ABC is an equilateral triangle of side length 30 cm. XY is parallel to BC , XP is parallel to AC and YQ is parallel to AB . If $(XY + XP + YQ)$ is 40 cm, then what is PQ equal to?



- (a) 5 cm (b) 12 cm
(c) 15 cm (d) None of these
- 38.** O is any point on the bisector of the acute angle $\angle XYZ$. The line OP is parallel to ZY . Then, $\triangle YPO$ is
- (a) scalene
(b) isosceles but not right angled
(c) equilateral
(d) right angled and isosceles
- 39.** The median BD of the $\triangle ABC$ meets AC at D . If $BD = \frac{1}{2} AC$, then which one of the following is correct?
- (a) $\angle ACB = 1$ right angle (b) $\angle BAC = 1$ right angle
(c) $\angle ABC = 1$ right angle (d) None of these
- 40.** ABC is a triangle, X is a point outside the $\triangle ABC$ such that $CD = CX$, where D is the point of intersection of BC and AX and $\angle BAX = \angle XAC$. Which one of the following is correct?
- (a) $\triangle ABD$ and $\triangle ACX$ are similar (b) $\angle ABD < \angle ACD$
(c) $AC = CX$ (d) $\angle ADB > \angle DXC$

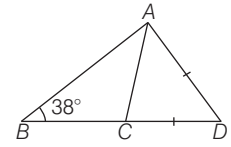


- 41.** In the figure given, $\angle ABD = \angle PQD = \angle CDQ = \frac{\pi}{2}$. If $AB = x$, $PQ = z$ and $CD = y$, then which one of the following is correct?



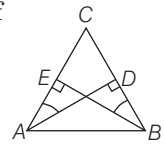
- (a) $\frac{1}{x} + \frac{1}{y} = \frac{1}{z}$
(b) $\frac{1}{x} + \frac{1}{z} = \frac{1}{y}$
(c) $\frac{1}{z} + \frac{1}{y} = \frac{1}{x}$
(d) $\frac{1}{x} + \frac{1}{y} = \frac{2}{z}$
- 42.** $\triangle PQR$ is right angled at Q , $PR = 5$ cm and $QR = 4$ cm. If the lengths of sides of another $\triangle ABC$ are 3 cm, 4 cm and 5 cm, then which one of the following is correct?
- (a) Area of $\triangle PQR$ is double that of $\triangle ABC$
(b) Area of $\triangle ABC$ is double that of $\triangle PQR$
(c) $\angle B = \frac{\angle Q}{2}$
(d) Both triangles are congruent

- 43.** In the figure, $\angle B = 38^\circ$, $AC = BC$ and $AD = CD$. What is the value $\angle D$?



- (a) 26° (b) 28°
(c) 38° (d) 52°
- 44.** Consider the following statements in respect of any triangle.
- I. The three medians of a triangle divide it into six triangles of equal area.
II. The perimeter of a triangle is greater than the sum of the lengths of its three medians.
- Which of the statement(s) given above is/are correct?
- (a) Only I (b) Only II (c) Both I and II (d) Neither I nor II

- 45.** Consider the following in respect of the given figure :



- I. $\triangle DAC \sim \triangle EBC$
II. $CA/CB = CD/CE$
III. $AD/BE = CD/CE$

Which of the statement(s) given above is/are correct?

- (a) II and III (b) I and II (c) I and III (d) All of these
- 46.** Consider the following statements :
- I. Let PQR be a triangle in which $PQ = 3$ cm, $QR = 4$ cm and $RP = 5$ cm. If D is a point in the plane of the $\triangle PQR$ such that D is either outside it or inside it, then

$$DP + DQ + DR > 6 \text{ cm}$$

II. PQR is a right angled triangle.

Which one of the following is correct in respect of the above two statements?

- (a) Both statements I and II are individually true and statement II is the correct explanation of statement I
(b) Both statements I and II are individually true but statement II is not the correct explanation of statement I
(c) Statement I is true and statement II is false
(d) Statement I is false and statement II is true

- 47.** I. Let LMN be a triangle. Let P, Q be the mid-points of the sides LM, LN , respectively. If $PQ^2 = MP^2 + NQ^2$, then LMN is a right angled triangle at L .

II. If in a $\triangle ABC$, $AB^2 > BC^2 + CA^2$, then $\angle ACB$ is obtuse.

Which of the following is correct in the light of the above statements?

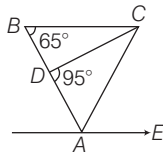
- (a) Both statements I and II are individually true and statement II is the correct explanation of statement I
(b) Both statements I and II are individually true but statement II is not the correct explanation of statement I
(c) Statement I is true and statement II is false
(d) Statement I is false and statement II is true

PREVIOUS YEARS' QUESTIONS

48. In a $\triangle ABC$, $\frac{1}{2}\angle A + \frac{1}{3}\angle C + \frac{1}{2}\angle B = 80^\circ$, then what is the value of $\angle C$? ☑ 2012 II
 (a) 35° (b) 40° (c) 60° (d) 70°

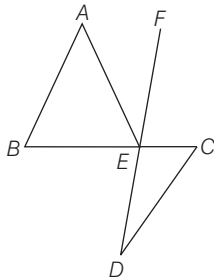
49. In a $\triangle ABC$, side AB is extended beyond B , side BC beyond C and side CA beyond A . What is the sum of the three exterior angles? ☑ 2012 II
 (a) 270° (b) 305° (c) 360° (d) 540°

50. In the given figure, ABC is a triangle, BC is parallel to AE . If $BC = AC$, then what is the value of $\angle CAE$? ☑ 2012 II



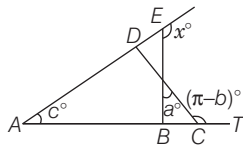
- (a) 20° (b) 30° (c) 40° (d) 50°

51. In the given figure, AB is parallel to CD . $\angle ABC = 65^\circ$, $\angle CDE = 15^\circ$ and $AB = AE$. What is the value of $\angle AEF$? ☑ 2012 II



- (a) 30° (b) 35° (c) 40° (d) 45°

52. The angles x° , a° , c° and $(\pi - b)^\circ$ are indicated in the figure given below : ☑ 2012 II



Which one of the following is correct? ☑ 2012 II

- (a) $x^\circ = a^\circ + c^\circ - b^\circ$ (b) $x^\circ = b^\circ - a^\circ - c^\circ$
 (c) $x^\circ = a^\circ + b^\circ + c^\circ$ (d) $x^\circ = a^\circ - b^\circ - c^\circ$

53. Consider the following statements:

- I. If G is the centroid of $\triangle ABC$, then $GA = GB = GC$.
 II. If H is the orthocentre of $\triangle ABC$, then $HA = HB = HC$.

Which of the statement(s) given above is are correct? ☑ 2013 II

- (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II

54. I. Let the side DE of a $\triangle DEF$ be divided at S , so that $\frac{DS}{SE} = \frac{1}{\sqrt{2}}$. If a line through S parallel to EF meets DF at T , then the area of $\triangle DEF$ is twice the area of the $\triangle DST$.

- II. The areas of the similar triangles are proportional to the squares of the corresponding sides.

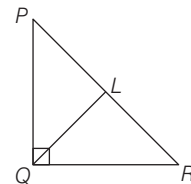
Which one of the following is correct in respect of the above statements? ☑ 2013 II

- (a) Both statements I and II are true and statement II is the correct explanation of statement I
 (b) Both statements I and II are true but statement II is not the correct explanation of statement I
 (c) Statement I is true but statement II is false
 (d) Statement II is true but statement I is false

55. Let ABC be a triangle with $AB = 3$ cm and $AC = 5$ cm. If AD is a median drawn from the vertex A to the side BC , then which one of the following is correct? ☑ 2013 II

- (a) AD is always greater than 4 cm but less than 5 cm
 (b) AD is always greater than 5 cm
 (c) AD is always less than 4 cm
 (d) None of the above

56. In the figure given above, $\angle PQR = 90^\circ$ and QL is a median, $PQ = 5$ cm and $QR = 12$ cm. Then, QL is equal to ☑ 2013 II



- (a) 5 cm (b) 5.5 cm (c) 6 cm (d) 6.5 cm

57. In a $\triangle ABC$, $\angle BCA = 90^\circ$ and CD is perpendicular to AB . If $AD = 4$ cm and $BD = 9$ cm, then the value of DC will be ☑ 2013 II

- (a) $\sqrt{18}$ cm (b) $\sqrt{20}$ cm (c) $\sqrt{65}$ cm (d) 6 cm

58. ABC is a triangle, where $BC = 2AB$, $\angle C = 30^\circ$ and $\angle A = 90^\circ$. The magnitude of the side AC is ☑ 2013 II

- (a) $\frac{2BC}{3}$ (b) $\frac{3BC}{4}$ (c) $\frac{BC}{\sqrt{3}}$ (d) $\frac{\sqrt{3} BC}{2}$

59. Let ABC be an equilateral triangle. If the side BC is produced to the point D so that $BC = 2CD$, then AD^2 is equal to ☑ 2013 II

- (a) $3CD^2$ (b) $4CD^2$ (c) $5CD^2$ (d) $7CD^2$

60. ABC is a right angled triangle such that $AB = a - b$, $BC = a$ and $CA = a + b$. D is a point on BC such that $BD = AB$. The ratio of $BD : DC$ for any value of a and b is given by ☑ 2013 II

- (a) 3 : 2 (b) 4 : 3 (c) 5 : 4 (d) 3 : 1